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DIFFERENCE-IN-DIFFERENCES WITH VARIATION IN TREATMENT TIMING

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extend recent research on DD models with heterogeneous effects.⁷ Assuming equal counterfactual trends, Abraham and Sun (2018), Borusyak and Jaravel (2017), and de Chaisemartin and D’Haultfœuille (2018b) show that two-way fixed effects DD yields an average of treatment effects across all groups and times, some of which may have negative weights. My results show how these weights arise from differences in timing and thus treatment variances, facilitating a connection between models of treatment allocation and the interpretation of DD estimates.⁸ I also explain why the negative weights occur: when already-treated units act as controls, *changes* in their treatment effects over time get subtracted from the DD estimate. This negative weighting only arises when treatment effects vary over time, in which case it typically biases regression DD estimates away from the sign of the true treatment effect. This does not imply a failure of the underlying *design*, but it does caution against the use of a single-coefficient two-way fixed effects specification to summarize time-varying effects.

I also show that because regression DD uses group sizes and treatment variances to weight up simple estimates that each rely on common trends between two groups, its identifying assumption is a variance-weighted version of common trends between all groups. The extent to which a group’s differential trend biases the overall estimate equals the difference between how much weight it gets when it acts as the treatment group and how much weight it gets when it acts as the control group. When the earliest and/or latest treated units have low treatment variance, they can get *more* weight as controls than treatments. In designs without untreated units they always

⁷ Early research in this area made specific observations about stylized specifications such as models with no unit fixed effects (Bitler, Gelbach, and Hoynes 2003), or it provided simulation evidence (Meer and West 2013).

⁸ Related results on the weighting of heterogeneous treatment effects does not provide this intuition. Abraham and Sun (2018, p 9) describe the weights in a DD estimate with constant treatment effects as “residual[s] from predicting treatment status, $D_{i,t}$ with unit and time fixed effects.” de Chaisemartin and D’Haultfœuille (2018b, p 7) and Borusyak and Jaravel (2017) describe these same weights as coming from an auxiliary regression and Borusyak and Jaravel (2017, p 10-11) note that “a general characterization of [the weights] does not seem feasible.” Athey and Imbens (2018) also decompose the DD estimator and develop design-based inference methods for this setting. Strezhnev (2018) expresses $\hat{\beta}^{DD}$ as an unweighted average of DD-type terms across pairs of observations and periods.